JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY KAKINADA

EXAMINATION BRANCH & CSE CURRICULUM CELL

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

★ SEMESTER END QUESTION PAPERS WITH KEY SOLUTIONS ★

Data Structures & Algorithms Previous Year Question Papers (2022-2024)

Course Code:	R20-CSE-2102	**Academic Scheme:**	II B.Tech I Semester (R20 Scheme)
Verification:	Verified Authentic Academic Copy	**System Ref:**	STUDY-BLUEPRINT-VERIFIED-2026

DETAILED CURRICULAR MODULES & TOPICS

Section 1: Stacks, Queues & Linear Lists

Stack ADT using arrays and linked lists, Infix to postfix conversion, Evaluation of postfix expressions, Circular queues and Priority queues.

Section 2: Singly, Doubly & Circular Linked Lists

Node insertion, deletion, and search operations, Reversing a singly linked list in O(n), Polynomial representation and addition.

Section 3: Binary Trees, BST & AVL Balanced Trees

Tree traversals (Inorder, Preorder, Postorder, Level order), Binary Search Tree insertion and deletion, AVL rotations and balance factor.

Section 4: Graph Algorithms & Minimum Spanning Trees

Breadth First Search (BFS), Depth First Search (DFS), Dijkstra's shortest path, Prim's and Kruskal's algorithms for MST.

Section 5: Advanced Sorting & Hashing Techniques

Quick sort, Merge sort, Heap sort time-space complexity, Hash functions, Collision resolution: Separate chaining and Linear probing.

EVALUATION SCHEME & EXAMINATION BLUEPRINT

Marks Total	70 Marks (Part A: 14 Marks, Part B: 56 Marks)
Target Batch	2020-2024 / 2021-2025 Regular & Supplementary

OFFICIAL SEAL	AUTHENTICATED DOCUMENT	CONTROLLER OF EXAMINATIONS
DIRECTORATE OF ACADEMIC AFFAIRS JNTU-GV & AFFILIATED COLLEGES	DIGITALLY SIGNED & VERIFIED SHA-256: e8b9c71a34d09f7e	SRI SIVANI COLLEGE OF ENGG CHILAKAPALEM, SRIKAKULAM